Controlling the RGB Cooling Fan

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- Controlling the Fan (On/Off)
- Controlling Lighting Effects

> [!NOTE]
> The RDK motherboard comes factory-equipped with a temperature control board, and the RGB lighting and fan will automatically turn on at boot.

Source Path: `/home/sunrise/software/oled_yahboom`

Controlling the Fan (On/Off)

```
236 if __name__ == "__main__":
237     try:
238         oled_clear = False
239         oled_debug = False
240         state = False
241         if len(sys.argv) > 1:
242             if str(sys.argv[1]) == "clear":
243                 oled_clear = True
244             elif str(sys.argv[1]) == "debug":
245                 oled_debug = True
246         oled = OLED(clear=oled_clear, debug=oled_debug)
247         try:
248             import smbus
249             bus = smbus.SMBus(5)
250             if not oled_clear:
251                 start = 1    # 风扇开关
252                 bus.write_byte_data(0x0d, 0x08, start)
253                 time.sleep(.05)
254                 effect = 4   # 灯光效果1-4
255                 bus.write_byte_data(0x0d, 0x04, effect)
256                 time.sleep(.05)
257         except:
258             pass
259
```

To turn on the fan: Set `start` = [1-9]. Note that `start = 1` sets the maximum speed, while for `start = 2~9`, higher values correspond to faster fan speeds.

```
start = 1
bus.write_byte_data(0x0d, 0x08, start)
time.sleep(.05)
```

To turn off the fan:

```
bus.write_byte_data(0x0d, 0x08, 0x00)
time.sleep(.05)
```

Controlling Lighting Effects

To enable lighting effects: Set `effect` = [0-4] to select the desired effect:

- `0`: Flowing Light
- `1`: Breathing Light
- `2`: Chasing Light / Marquee
- `3`: Rainbow Light
- `4`: Dazzling Light

```
effect = 2
bus.write_byte_data(0x0d, 0x04, effect)
time.sleep(.05)
```

To turn off the lights:

```
bus.write_byte_data(0x0d, 0x07, 0x00)
time.sleep(.05)
```